

Summary Report

1. Project Overview Machine learning is increasingly accelerating materials discovery by serving as a high-speed surrogate for computationally expensive density functional theory (DFT) simulations. Specifically, predicting fundamental properties from composition alone—known as structure-agnostic modeling—is critical for discovering novel hypothetical materials whose 3D atomic coordinates are not yet known.

This project focuses on developing predictive models for several key materials properties, including:

- **Formation Energy and Thermodynamic Stability (Hull Distance):** These metrics are vital for determining whether a newly proposed material is thermodynamically stable enough to be synthesized in a lab or if it will decompose into more stable phases.
- **Space Group:** Understanding crystal symmetry is highly useful for analyzing physical properties and serves as a critical first step in determining the atomic coordinates of a material.
- **Volume per Atom:** This property serves as a fundamental proxy for density, packing efficiency, and mechanical properties.

By leveraging large, high-throughput materials databases, this project evaluates how different model architectures and feature representations perform across varied datasets and target properties.

2. Datasets Used The project utilizes data sourced from two major density functional theory (DFT) calculated repositories: the Materials Project (MP) and the AFLOW database.

- **Materials Project (MP) Dataset:**
 - *Size and Space Groups:* The dataset size varies depending on the specific modeling task. For space group classification, the MP dataset contains 73,704 samples covering 99 unique space groups. For the volume per atom regression task, a larger cleaned dataset containing 174,590 instances was utilized.
 - *Features:* The data includes parsed chemical formulas, target physical properties, and a comprehensive suite of Magpie descriptors that capture elemental statistics (minimum, maximum, range, mean, and average deviation).
- **AFLOW Dataset:**
 - *Size and Space Groups:* The raw AFLOW dataset is massive, initially comprising 2,981,873 instances and representing a much broader diversity of 219 unique space groups.
 - *Preprocessing and Sampling:* After dropping instances with missing space groups, 2,981,344 rows remained. To manage computational memory constraints during the training of Random Forest classifiers, the data was randomly sampled down to 300,000 rows. For neural network multi-task learning, specific targets

like `Egap` and `energy_cell` were mapped and renamed to `formation_energy` and `hull_distance`.

- **Key Differences & Cleaning:** AFLOW offers a much larger scale than the MP dataset, providing a more rigorous test of model scalability. A critical preprocessing step applied across both datasets was the identification and removal of "data leakage" columns (such as `MagpieData mean SpaceGroupNumber`) that would have unfairly exposed the target variable to the models during training. Additionally, rare space group classes containing only one sample were filtered out to ensure valid train/test stratification.

3. Feature Engineering Transforming chemical formulas into machine-readable numerical inputs was a primary focus of the data pipeline, emphasizing composition-based representations.

- **Formula Parsing and Fractional Encodings:** Chemical formulas were systematically parsed into two distinct arrays: `elements` (which maps atomic symbols to integer atomic numbers 1 through 118) and `fractions` (the normalized stoichiometric proportions of each element in the compound). String representations of these arrays were safely evaluated and converted to PyTorch tensors for neural network consumption.
- **Elemental Embeddings:** In the neural network architectures (like `CompositionMTL` and `VolumePerAtomUQ`), the parsed atomic numbers are fed into an `nn.Embedding` layer to learn high-dimensional dense vector representations. These embeddings are then multiplied by their respective fractional weights and summed along the dimension to create a single, unified composition vector for the entire material.
- **Physical Descriptors:** For the Random Forest models, Magpie features provided a rich set of physicochemical descriptors based on elemental statistics. Additional domain-knowledge feature engineering was applied by calculating custom columns: `radius_range` (the difference between the maximum and minimum covalent radii), `valence_std` (average deviation in valence electrons), and `packing_proxy` (the ratio of the true volume per atom to the Magpie mean ground state volume). Furthermore, for certain neural network tasks, explicit physical properties (like mean atomic radius and mean electronegativity) were extracted and passed through a dedicated physics feature encoder network.
- **Categorical Encodings:** Categorical structural identifiers available in the AFLOW data, such as `Pearson_symbol_relax`, were one-hot encoded using dummy variables to allow the models to process them numerically.

4. Model Architectures To address the diverse set of property predictions, this project employed both ensemble machine learning techniques and specialized deep learning architectures.

- **Random Forest Ensembles (Space Group Classification):** Predicting the space group of a material was treated as a highly imbalanced multi-class classification problem. To tackle this, Random Forest classifiers were utilized. The models were configured with `n_estimators` ranging from 200 to 600, a `max_depth` of 20, and `min_samples_leaf` between 2 and 5. To address class imbalance across the 200+ space groups, a `class_weight="balanced"` strategy was applied, which dynamically adjusted weights inversely proportional to class frequencies. To improve robustness, predictions were generated using an ensemble of two independently initialized Random Forest models, where their output probabilities were averaged.
- **CompositionMTL (Multi-Task Learning Neural Network):** For predicting formation energy and thermodynamic stability (hull distance), a custom PyTorch model named `CompositionMTL` was designed. This architecture maps atomic numbers into dense, continuous vector spaces using an `nn.Embedding` layer (118 elements to 64 dimensions). The embeddings are passed through a shared Multi-Layer Perceptron (MLP) consisting of linear layers, ReLU activations, and a Dropout layer (0.3) for regularization. The processed embeddings are then multiplied by their fractional stoichiometric weights and summed across the composition to create a single context vector. This shared representation branches into three specialized output heads: `fe_head` (predicts continuous formation energy), `hull_head` (predicts continuous hull distance), and `stab_head` (uses a Sigmoid activation to perform binary classification of thermodynamic stability).
- **VolumePerAtomUQ (Probabilistic Neural Network):** Predicting the volume per atom utilized a separate dual-branch architecture named `VolumePerAtomUQ`. The first branch is a composition encoder identical in principle to the one used in `CompositionMTL`. The second branch is a *physics feature encoder* that processes five explicit physicochemical descriptors (e.g., mean atomic radius, mean electronegativity) through its own dense layers. The outputs of both encoders are concatenated and passed through a fusion layer. Crucially, the final output utilizes probabilistic heads: a `mu_head` to predict the expected volume per atom, and a `logvar_head` to predict the log variance. This design allows the model to output a target prediction alongside an estimate of aleatoric/epistemic uncertainty.

5. Training Procedures The training pipelines were heavily optimized for handling the complexities of materials datasets and probabilistic deep learning.

- **Data Splitting and Batching:** Datasets were strictly split to prevent data leakage, typically using an 80% train and 20% test split, with stratification applied to classification tasks to ensure rare space groups were present in both sets. For neural networks, 20% of the training pool was further separated into a validation set to monitor generalization.

Because chemical formulas contain varying numbers of elements, custom `collate_fn` functions were used in PyTorch DataLoaders to dynamically pad the element and fraction tensors with zeros to the maximum length within each specific batch.

- **Optimization and Loss Functions:** Neural networks were optimized using the Adam optimizer with a learning rate of $1e-3$. The multi-task `CompositionMTL` model minimized a combined loss function: Mean Squared Error (MSE) for the continuous formation energy and hull distance targets, and Binary Cross-Entropy (BCE) for the stability classification. The `VolumePerAtomUQ` model minimized a custom Gaussian Negative Log-Likelihood (NLL) loss, simultaneously optimizing the prediction `mu` and the variance `log_var`.
- **Epochs, Early Stopping, and Deep Ensembles:** Neural models were trained for 100 to 150 epochs. During training, the validation Mean Absolute Error (MAE) was continuously monitored, and the weights of the best-performing epoch were actively saved to disk to prevent overfitting. To capture epistemic uncertainty and improve predictive accuracy, the neural network training was wrapped in a Deep Ensemble loop, training up to 10 independent models with different initialization seeds. The final predictions were computed as the mean of the ensemble, with the standard deviation serving as the uncertainty bound.

6. Evaluation Metrics To rigorously evaluate model performance across both physical regressions and categorical classifications, a diverse suite of metrics was utilized:

- **Mean Absolute Error (MAE):** Used as the primary evaluation metric for continuous properties (formation energy, hull distance, and volume per atom). In materials science, reaching an MAE close to or below "chemical accuracy" is critical for validating whether a surrogate ML model can reliably filter candidate structures.
- **Top-\$k\$ Accuracy (Top-1, Top-3, Top-5):** Deployed for space group classification. Because many space groups share highly similar symmetry operations and datasets are imbalanced, Top-\$k\$ accuracy calculates whether the true space group was present within the model's top 1, 3, or 5 most confident predictions.
- **Precision, Recall, and F1-Score:** Leveraged for evaluating binary thermodynamic stability and space group classes. The F1-score (the harmonic mean of precision and recall) was prioritized over raw accuracy because it penalizes models that blindly predict the majority class.
- **ROC-AUC (Receiver Operating Characteristic - Area Under Curve):** Used specifically for the stability classification task (where stable materials have a hull distance of 0). An AUC close to 1.0 indicates that the model is highly capable of separating stable materials from unstable ones across all probability thresholds.
- **Uncertainty-Error Correlation and Coverage vs. Accuracy:** To evaluate the probabilistic outputs of the models, the correlation between the predicted uncertainty (standard deviation of the ensemble) and the actual absolute error was measured. Furthermore, Coverage vs. Accuracy metrics were plotted to demonstrate risk-controlled screening, proving that if the models are allowed to abstain from predicting highly uncertain materials, the overall MAE drops significantly.

7. Experimental Results The project conducted extensive evaluations across multiple target properties on both the Materials Project (MP) and AFLOW datasets. The models demonstrated varying degrees of success depending on the dataset scale, the complexity of the target property, and whether the task was framed as regression or classification.

Multi-Task Regression and Classification (Formation Energy & Stability): For the `CompositionMTL` neural network, which simultaneously predicted Formation Energy (FE), Hull Distance, and Thermodynamic Stability, the ensemble approach yielded highly robust results, particularly on the MP dataset.

- **MP Dataset:** The 10-model ensemble achieved a highly accurate FE Mean Absolute Error (MAE) of 0.097 eV/atom and a Hull Distance MAE of 0.075 eV/atom. For binary stability classification, the model demonstrated exceptional separability with an Accuracy of 0.898, an F1-score of 0.931, and a ROC-AUC of 0.956.
- **AFLOW Dataset:** Tested on a subset of the vast AFLOW database, the ensemble produced an FE MAE of 0.470 eV/atom and a significantly higher Hull Distance MAE of 29.578 eV/atom (due to `energy_cell` being mapped to the hull distance proxy). Stability classification remained strong with an Accuracy of 0.891, an F1-score of 0.798, and a ROC-AUC of 0.948.

Space Group Classification (Random Forest Ensembles): The Random Forest classifiers were evaluated using Top- k accuracy to account for the inherent difficulty of predicting exact symmetries from composition alone.

- **AFLOW Dataset (169 classes after filtering):** The ensemble achieved a raw Top-1 Accuracy of 0.682, a Top-3 Accuracy of 0.965, and a Top-5 Accuracy of 0.975.
- **MP Dataset (99 classes):** The raw ensemble yielded a Top-1 Accuracy of 0.656, Top-3 of 0.835, and Top-5 of 0.881. However, when predictions were systematically re-ranked by leveraging a secondary model that first predicted the broader crystal system, the performance surged significantly: Top-1 reached 0.774, Top-3 reached 0.895, and Top-5 hit 0.920.

Volume per Atom (Probabilistic Regression): The `VolumePerAtomUQ` model, trained exclusively on the MP dataset, achieved a final test MAE of 0.677 Å³/atom. Crucially, the model's probabilistic outputs demonstrated a strong Uncertainty-Error Correlation of 0.680, proving that the model accurately “knows what it doesn't know”.

8. Visualizations To provide transparency and deep physical insights into the model behaviors, several advanced visualization techniques were implemented:

- **t-SNE Projection of Learned Element Embeddings:** The 64-dimensional dense vectors learned by the neural network's embedding layer were projected into a 2D space using t-SNE. This visualization confirmed that the model implicitly learned the periodic table and fundamental chemical rules (grouping similar elements together) strictly from stoichiometry, without requiring handcrafted physical descriptors.

- **Density-Aware Parity Plots with Marginal Histograms:** Because standard scatter plots suffer from severe over-plotting with large test sets (~15,000 samples), Hexbin density plots were utilized. These plots, equipped with marginal distribution curves, proved that the vast majority of predictions were tightly concentrated along the ideal 1:1 diagonal, indicating highly localized errors rather than widespread dispersion.
- **Ensemble Predictions with Uncertainty Bounds:** Inspired by leading research, scatter plots were augmented with vertical error bars (± 1 standard deviation) derived from the Deep Ensemble. This highlighted the models' epistemic and aleatoric uncertainty. Corresponding **Coverage vs. Accuracy** curves demonstrated risk-controlled screening capabilities, showing that MAE drops significantly if the model abstains from predicting highly uncertain instances.
- **Element-Wise Error Heatmaps across the Periodic Table:** This visualization mapped the average MAE based on the constituent elements present in the compounds. It acts as a diagnostic tool, providing immediate transparency regarding which chemical spaces (e.g., specific transition metals) the model excels at and which are inherently harder to predict from composition alone.
- **Feature Importance and Class-wise Heatmaps:** For the Random Forest models, horizontal bar charts highlighted the top 20 most influential composition descriptors (such as `MagpieData mean Column` and `avg_dev NpValence`). Additionally, per-class Sensitivity and Specificity heatmaps were plotted for the space groups to diagnose imbalanced classification performance across specific symmetries.

9. Comparative Analysis Between Datasets Comparing the models trained on the Materials Project (MP) versus the AFLOW dataset reveals critical insights regarding the trade-offs between dataset scale, feature representation, and structural diversity.

- **Scale vs. Regression Complexity:** The raw AFLOW dataset is massive (~3 million instances) compared to the MP dataset. However, for continuous regression tasks like Formation Energy, the MP-trained models drastically outperformed the AFLOW models (0.097 eV/atom vs. 0.470 eV/atom). This suggests that simply scaling up dataset size does not necessarily improve structure-agnostic regression if the underlying dataset contains higher structural diversity, polymorphism, or noisier unrelaxed states that overwhelm the purely compositional signal.
- **Classification Resilience:** Despite the regression difficulties, binary classification for thermodynamic stability remained remarkably resilient across both datasets. The models achieved a ROC-AUC of 0.956 on MP and 0.948 on AFLOW. This implies that learning the decision boundary between "stable" and "unstable" from stoichiometry is a highly generalized capability that scales well regardless of database origin.
- **Space Group Prediction:** The AFLOW dataset provided a much richer distribution of space groups (169 valid classes after filtering vs. MP's 99). The raw Top-3 accuracy on AFLOW (0.965) exceeded the raw Top-3 on MP (0.835). This demonstrates that for highly multi-class, imbalanced categorization tasks, the sheer volume of data in AFLOW provided the Random Forest ensembles with the necessary statistical support to map complex stoichiometric ratios to probable crystal symmetries.

- **Target Definitions:** A notable divergence causing performance differences was the availability and naming of specific features. For instance, the MP dataset had clean `formation_energy_per_atom` and `energy_above_hull` targets. In contrast, AFLOW required mapping `Egap` and `energy_cell` to proxy targets, resulting in massive variance for the AFLOW hull distance predictions (MAE of ~29.5 eV/atom). This highlights that meticulous data curation and physically consistent target definitions are just as critical as model architecture.

10. Comparison With Existing Research To fully contextualize the achievements of this project, it is essential to benchmark the results against the current state-of-the-art materials informatics literature highlighted in the sources.

- **Structure-Aware Models (MEGNet and DeeperGATGNN):** Models like MEGNet and DeeperGATGNN leverage explicit 3D atomic coordinates and bond distances using crystal graph networks. MEGNet achieves an exceptional Formation Energy MAE of approximately 0.028 eV/atom, and DeeperGATGNN reaches ~0.032 eV/atom. Because our `CompositionMTL` model is strictly *structure-agnostic* (using only stoichiometry), our MAE of 0.097 eV/atom is naturally higher. However, our model bypasses the computational bottleneck of requiring pre-calculated DFT structures, making it dramatically faster for screening hypothetical materials. Furthermore, just as MEGNet utilized t-SNE projections to prove its model implicitly learned the periodic table from crystal graphs, our t-SNE analysis confirms that our multi-task MLP successfully learned these fundamental chemical relationships strictly from composition data.
- **Structure-Agnostic Models (CrabNet and Roost):** Our project aligns more closely with composition-only frameworks like CrabNet and Roost. CrabNet utilizes a computationally heavy Transformer self-attention mechanism to achieve an MAE of 0.077 eV/atom on the Materials Project (MP) dataset. Our multi-task MLP achieves a comparable MAE of 0.097 eV/atom while using a much more lightweight, scalable architecture. Additionally, while Roost emphasizes deep ensembles to estimate epistemic and aleatoric uncertainty (achieving 0.0241 eV/atom on OQMD), our project extends this probabilistic approach by simultaneously framing Thermodynamic Stability as a highly robust binary classification (F1-score of 0.931), which Roost does not explicitly screen for.
- **Space Group Prediction (Zhao et al. and Venkatraman et al.):** Prior research by Zhao et al. utilized Random Forests with Magpie features to predict space groups, achieving an F1 score of 0.650 for multiclass prediction on MP data. Our project builds on this foundation by employing ensemble Random Forests and utilizing domain-knowledge features (like `radius_range` and `valence_std`), alongside a hierarchical re-ranking strategy based on crystal systems, pushing top-3 accuracies to near 0.90.

11. Limitations of the Current Work While the models demonstrated strong predictive capabilities, several inherent limitations were identified during the development and evaluation phases:

- **Proxy Targets and Dataset Discrepancies:** A major limitation encountered with the AFLOW dataset was the lack of direct, clean targets for thermodynamic stability. To train the models, `Egap` and `energy_cell` had to be mapped as proxies for formation energy and hull distance. This proxy mapping resulted in a massive variance and an unacceptably high Hull Distance MAE (~ 29.5 eV/atom) for the AFLOW models.
- **Composition-Only Performance Ceiling:** Because the models rely exclusively on stoichiometry and elemental properties, they suffer from an inherent inability to distinguish between polymorphs—materials that share the exact same chemical formula but arrange into different 3D structures with vastly different properties. This creates a performance ceiling compared to structure-aware graph neural networks.
- **Extreme Class Imbalance:** Predicting space groups remains hindered by severe class imbalance. The AFLOW dataset originally contained 219 space groups, but many classes contained only a single sample, requiring aggressive filtering (down to 169 classes) to allow for valid train/test stratification. This scarcity prevents the model from reliably learning the complex stoichiometric ratios of rare symmetries.

12. Future Work To overcome the current limitations and push the boundaries of materials discovery, several future research directions are proposed:

- **Integration of Graph Neural Networks (GNNs):** For downstream tasks where hypothetical structures have been generated (e.g., via generative adversarial networks or structural relaxation), transitioning from MLP architectures to deep graph neural networks (such as ALIGNN or DeeperGATGNN) would significantly improve regression accuracy by capturing 3D atomic bonding environments.
- **Transfer Learning Across Databases:** Future iterations should leverage transfer learning to address data scarcity. For example, elemental embeddings could be pre-trained on massive datasets (like OQMD or the 3-million sample AFLOW dataset) and then fine-tuned on the much cleaner, meticulously curated MP dataset.
- **Enhanced Data Curation:** Future pipelines must prioritize rigorous data curation over raw dataset scale. Fixing the proxy target variance in AFLOW by calculating strict energy-above-hull metrics will be critical to making AFLOW-based thermodynamic predictions viable.

13. Conclusion This research project successfully developed and evaluated a suite of machine learning models for the rapid, structure-agnostic screening of inorganic materials. By leveraging dense elemental embeddings and multi-task learning (`CompositionMTL`), the models achieved high accuracy in predicting continuous properties like formation energy alongside highly robust binary classifications for thermodynamic stability.

Furthermore, the implementation of probabilistic neural networks (`VolumePerAtomUQ`) and Deep Ensembles provided critical uncertainty estimations, proving that risk-controlled screening can effectively filter out unreliable predictions. While the comparative analysis revealed that massive dataset scale (AFLOW) cannot compensate for noisy target definitions compared to smaller, high-quality datasets (Materials Project), the overall framework demonstrates that

stoichiometry alone contains a profound amount of chemical intuition. Ultimately, these lightweight surrogate models offer a highly efficient, scalable first step in computational materials discovery pipelines, drastically reducing the search space before computationally expensive DFT simulations are required.